universität wien
Faculty of Psychology

Lernen unter COVID-19-Bedingungen

Students (longitudinal study)

Method Report

1. Introduction

The competence for self-regulated learning (SRL) are considered important determinants of learning achievements, well-being, and ultimately health (e.g., Schober et al., 2013; Wirth & Leutner, 2008). In the wake of the COVID-19 pandemic, students worldwide had to switch to home learning more or less unprepared and faced a fundamentally altered situation where self-regulated learning has become a necessity. The present longitudinal study aims at investigating the relevance of SRL for well-being and learning success, both quantitatively and qualitatively, in a large sample consisting of Austrian university students, at three measurement points in the first half of 2020.

2. Study Description

2.1. Title and version

Lernen unter COVID-19-Bedingungen: Studierende (Längsschnittstudie)
Learning under COVID-19 conditions: students, V1.0 (longitudinal study)

2.2. Principal Investigators

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2.3. Funding/Acknowledgments

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2.4. Fieldwork agency

The fieldwork of the study “Lernen unter COVID-19-Bedingungen” was managed and handled internally.

2.5. Topics/Keywords

- Sociodemographic information: age, sex, gender-role orientation, education
- Situation at home: employment, study time, care-giving duties
- Crisis perception: perceived challenges, information distribution, university and classes attendance,
- Effects of the crisis: changes in well-being, learning, motivation, social relatedness, competence, autonomy, and self-regulated learning

- Learning: basic needs (social relatedness, competence, autonomy), self-regulated learning, learning goals, learning time, learning environment, metacognition, procrastination, flow, perseverance, learning motivation
- Education: achievement, satisfaction, importance of grades
- Online-education: digital learning, digital teaching, student-teacher-contact, time spent on studying

3. Study Design

3.1. Sampling

The sample of the study "Lernen unter COVID-19-Bedingungen" consisted of Austrian students attending higher education with access to the internet. The students were invited to participate in the first measurement point by calls in radio and television programs. Moreover, the link to and a recommendation to participate in the survey was published by the Federal Ministry of Education, Science, and Research on their website. The link with a respective invitation to the study was also placed on the survey's website. In order to participate in the study, the respondents had to agree to the informed consent, be Austrian residents, and attend higher education. Participants were asked to give their email address if they would like to be contacted with the link for the questionnaire in measurement point 2 and 3. Only those that gave their email and consent to be contacted in the following measurement points were contacted. Only those that filled in the questionnaire using their personalized invitation are included in the longitudinal part of the project. It is important to note that the sample cannot be considered representative for the Austrian population as many students that not have easy access to the internet and/or devices were not able to participate in measurement point 1 and then to be invited to measurement point 2 and 3. Therefore, risk groups may be underrepresented in the current sample.

3.2. Fieldwork

The overall data collection was conducted in the time from 07 April 2020 to 29 June 2020.

- Wave 1: 26 March 2020 – 24 April 2020
- Wave 2: 24 April 2020 – 12 May 2020
- Wave 3: 08 June 2020 – 29 June 2020

4. Data Cleaning

4.1. General data cleaning procedure

Cleaning procedure was the same for all three data sets collected in three measurement points and contained two steps:

4.1.1. Initial data cleaning

As participating students answered an online questionnaire via Unipark (Questback, 2007), data was exported from unipark in .csv format and imported to SPSS. Variables names were replaced. All missing values were defined as -99. Invalid zero values or values of -77 were replaced with -99. Participants who gave consent and had answering time (variable duration) larger than zero were selected for a new SPSS file.

4.1.2. Data cleaning

As unipark assigns -1 to variable duration in cases where the break was made during the answering the questionnaire, all valid -1 cases were replaced with (+) 9999999 value. For all open-ended questions that needed to be cleaned a new variable was made with suffix .clean in the variable name and there all non-integer values were replaced with a valid integer value (e.g., mean values of stated inputs, see the rules for cleaning open-ended questions below), all string values were converted to numeric (see below) and all invalid inputs were replaced with -99 (see example below). Participants with 100% missings on main outcome variables (positive affect, social relatedness, competence, autonomy, SRL scales and procrastination) were excluded. Minimum possible duration was tested by filling in the questionnaire as fast as possible while reading all items. All cases above the duration threshold and within a defined age-range (based on data from Statistics Austria) were selected for a new SPSS file.

Only in measurement point 1 participants were asked questions related to sociodemographic data. Participants were able to enter a textual answer to question on study cycle and type of university they attend. Those open-ended answers were checked to make sure that participants attend an institution of higher education. Participant that wrote secondary school type or name of a college or course that are not consider as higher education in Austria were excluded from the data set.

4.1.3. Missing values

All empty spaces were replaced with -99. Missing values that were represented by 0 in the raw data set are replaced by -99 on all variables where 0 is not a valid value. Missing values represented with -77 in unipark were replaced with -99 as well.

4.1.4. Filters

Data records of participants who did not give their consent are deleted.

A fastest possible time for answering the questionnaire is determined diligently. Variable duration of answers with duration below the fastest possible time is set to 9999999.

Data records with 100% missings on main outcome variables are deleted.

4.1.5. Answers to open questions

Answers to answering option that allowed open-ended answer were screened for invalid responses (see Data cleaning above). For time variables the numerical value is extracted from the string input and then categorized (as.employ.norm.hours, as.employ.curr.hours, as.study.norm.rec, as.study.curr.rec).

4.1.6. Multiple choice / single choice

All variables with prefix **as.uni** in measurement point 1 portray the answers to the university type(s) attended by each participant, as the question "An welcher Hochschule studieren Sie?" in measurement point 1 was posed in multiple choice format.

4.2. Anonymisation procedure

4.2.1. Direct identifiers

Variables containing direct identifiers are deleted (**as.email**).

4.2.2. Open questions

In measurement point 1 the following open variables are deleted without recoding or standardizing the answers:

- **as.major** ("Welche Studienrichtung studieren Sie?")
- **as.concerns** ("Worüber machen Sie sich *derzeit* die größten Sorgen?")
- **as.pos** ("Was empfinden Sie *derzeit* als besonders positiv? ")
- **as.chall.study** ("Was sind für Sie *derzeit* die größten Herausforderungen beim Studieren?")
- **as.coping** ("Wie gehen Sie mit diesen Herausforderungen beim Studieren um?")
- **as.coping.succ** ("Wie gut gelingt Ihnen das?")
- **as.study.succ** ("Was gelingt Ihnen in Bezug auf das Studieren *derzeit* besonders gut?")
- **as.support** ("Welche Unterstützungen würden Sie *derzeit* beim Studieren brauchen?").

In measurement point 2 the following open variables are deleted without recoding or standardizing the answers:

- **bls.more.info** ("Worüber würden Sie mehr Information brauchen?")
- **bls.studya.better** ("Wenn sich etwas *verbessert* hat: Was ist das genau?")
- **bls.studya.worse** ("Wenn sich etwas *verschlechtert* hat: Was ist das genau?")
- **bls.dont.miss** ("Was vermissen Sie gar nicht?")
- **bls.miss.most** ("Was vermissen Sie am meisten?")

In measurement point 3 the following open variables are deleted without recoding or standardizing the answers:

- **cqs.classes.no.attendance.other** ("Warum besuchen Sie die angebotenen Präsenzeinheiten nicht? Andere")
- **cqs.no.exam.other** ("Warum haben Sie sich bisher zu keiner Online-Prüfung angemeldet?")
- **cqs.exam.written.other** ("Falls die Prüfung in einem schriftlichen Online-Format stattgefunden hat, in welchem? Andere")
- **cqs.exam.pos** ("Was finden Sie am Format der Online-Prüfung besonders positiv?")
- **cqs.exam.neg** ("Was finden Sie am Format der Online-Prüfung besonders negativ?")

- cqs.no.online.exam.taken.other (“Warum haben Sie nicht an der Online-Prüfung teilgenommen?”)
- cqs.studya.better (“Wenn sich etwas *verbessert* hat: Was ist das genau?”)
- cqs.studya.worse (“Wenn sich etwas *verschlechtert* hat: Was ist das genau?”)
- cqs.dont.miss (“Was vermissen Sie gar nicht?”)
- cqs.miss.most (“Was vermissen Sie am meisten?”)
- cqs.hl.experience (“Was nehmen Sie aus der Zeit des Home-Learnings für später mit?”)
- cqs.hl.hobbies (“Haben Sie während der Zeit zu Hause neue Dinge für sich entdeckt (z.B. neue Hobbies, Sportarten, …)?”)
- cqs.future.worries (“Wenn Sie an den kommenden Herbst denken, worüber machen Sie sich am meisten Gedanken?”)

4.2.3. Open questions containing numeric values

Answers to questions requiring numeric values are recoded according to the description in section "Data cleaning". Original (string) answers are deleted. Furthermore, the cleaned variables containing the extracted numerical values are deleted for anonymization reasons in case they contain sensitive data (as.employ.norm.hours, as.employ.curr.hours).

4.2.4. Age variable

Data containing the age of the participants are in measurement point 1 categorized by generating age groups with at least 20 observations each.

4.2.5. University type variables

In measurement point 1 values containing "Fernfachhochschule" or "Fernuniversität" are merged to one category ("Fernfachhochschule/Fernuniversität").

This provides groups with at least 20 observations.

4.3. Categorizing variables

Variables containing numeric values in connection with time and numbers of people were categorized into suitable categories that allow analyses among different points of measurement.

4.4. Further deletions

4.4.1. Consent variable

Variables containing the participants' agreement on processing the filled in data for scientific purposes are deleted in measurement point 1 (as.consent, as.consent2) as all included participants gave their consent for processing the

filled in data for scientific purposes. One variable containing the participants' intention to participate in upcoming measurement points is deleted in measurement point 1 (as.continue).

4.4.2. Time method identifiers

Variables referring to questions that help decide if the participant is selected for the cross-sectional or the longitudinal study are deleted (bqs.first.mail, cqs.first.mail).

4.4.3. Filter variables

One filter variable (filter_$) originating from the cleaning procedure is deleted in each measurement point.

4.4.4. System variables

System variables that contain information of the duration of answering separate pages are deleted.

Measurement point 1:

- browser (information on the web browser used)
- referer (website containing the link leading to the survey)
- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)
- rts5442957, rts5442958, rts5444108, rts5444131, rts5444193, rts5444194, rts5444264, rts5444306, rts5444309, rts5444311, rts5453777, rts5453778, rts5455741, rts5455804, rts5455897 (relative timestamps for pages)

Measurement point 2:

- browser (information on the web browser used)
- referer (website containing the link leading to the survey)
- rts4286862, rts4286865, rts4286868, rts4286869, rts4286870, rts4286871, rts4286873, rts4286874, rts4290187, rts4290188, rts4290189, rts4290834 (relative timestamps for pages)

Measurement point 3:

- browser (information on the web browser used)
- referer (website containing the link leading to the survey)
- date_of_first_mail (date of first e-mail)
- datetime (date of first access)
- date_of_last_access (date of last access)

- rts4415819, rts4415822, rts4415826, rts4415827, rts4415828, rts4415829, rts4415830, rts4415831, rts4415832, rts4415833, rts4415836, rts4415841, rts4415859, rts4415863, rts4415868, rts4415873, rts4415876, rts4415877, rts4415930, rts4416523, rts4416756 (relative timestamps for pages)

4.5. Increasing reusability

To increase the reusability of the data set variables are rearranged into an appropriate order. Variable names are renamed to get more meaningful names.